The Schuler Bridge: Navigation Timescale, Light-Time, and the First-Order Clock Rate

Light-Meter Physics

Abstract

In an ideal spherical-Earth model, the Schuler angular frequency is set by the ratio of surface gravity to Earth radius. Expressing that radius as its vacuum light-time produces two algebraic identities: one gives the acceleration scale g/c, and a second gives the dimensionless compactness gR/c^2. When g is the Newtonian surface acceleration GM/R^2, this compactness is the leading weak-field excess of the inverse static-clock rate $\delta\tau \equiv \mathrm{d}t/\mathrm{d}\tau$ relative to time at infinity. A consistent Earth example yields a Schuler period of about 84.35 minutes and a first-order surface-clock lag of about 60.1 microseconds per reference day. The connection is dimensional and algebraic; it does not mean that navigation dynamics cause gravitational time dilation, or that any physical signal crosses the Earth. The Schuler model is Newtonian, while the clock-rate comparison is retained only to first order in $GM/(Rc^2)$. The paper also distinguishes the cycle frequency from angular frequency and the ideal-sphere acceleration from the conventional standard-gravity value.

Keywords: Schuler tuning, inertial navigation, gravitational time dilation, clock rate, weak field

1 Introduction

Schuler tuning gives a characteristic oscillation period for a simplified inertial-navigation error mode near a spherical Earth. The same scale can be represented using the Earth radius as a vacuum light-time. Combining the navigation frequency with that time converts the length scale into a dimensionless quantity that also appears in the weak-field comparison between a static surface clock and a clock at infinity.

The purpose of this manuscript is to show that bridge with explicit units and a controlled approximation. The navigation result is the standard Schuler relation, not a new dynamical law [1, 2]. The clock comparison begins with the static Schwarzschild lapse [3], but only its first-order weak-field limit is used in the bridge. The algebra does not establish a causal connection between inertial navigation and gravity; it shows that familiar scales can be rearranged into the same dimensionless ratio.

The model is an ideal, spherical, nonrotating Earth with mass M and radius R. The reference time t is normalized to proper time at infinity, and τ is the proper time of a static clock at the surface. Earth's rotation, flattening, local gravity variation, other bodies, and the geoid are excluded. These assumptions make the result a transparent model comparison rather than a prediction for a real clock-transfer system.

2 Schuler frequency and the ideal Earth

Let g_{N} denote the Newtonian surface-gravity magnitude of the ideal sphere:

$$g_{\mathrm{N}} \equiv \frac{GM}{R^2}. \tag{1}$$

The Schuler mode of a simplified, unforced, linearized inertial-navigation model has angular frequency

$$\omega_{\mathrm{S}}^2 = \frac{g_{\mathrm{N}}}{R}, \qquad \ddot{q} + \omega_{\mathrm{S}}^2 q = 0, \tag{2}$$

where q denotes the small error coordinate of that mode. This is not a claim that every navigation error follows a single sinusoid; sensor errors, damping, rotation, and other model terms can alter the complete error dynamics. The same frequency is often represented by an equivalent pendulum of length R, whose ideal period is $2\pi\sqrt{R/g_{\mathrm{N}}}$. No physical Earth-length pendulum is implied.

Navigation papers commonly state the Schuler frequency in cycles per second. Define

$$f_{\mathrm{S}} \equiv \frac{\omega_{\mathrm{S}}}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{g_{\mathrm{N}}}{R}}, \qquad T_{\mathrm{S}} \equiv \frac{1}{f_{\mathrm{S}}} = 2\pi\sqrt{\frac{R}{g_{\mathrm{N}}}}. \tag{3}$$

The factor of 2π distinguishes a cyclic frequency from an angular frequency measured in radians per second. Retaining that distinction is essential when the frequency is squared below.

3 Earth radius expressed as a light-time

Define the radius light-time

$$t_R \equiv \frac{R}{c}, \tag{4}$$

where c is the vacuum speed of light. This is a unit conversion from length to time. It does not describe a photon traveling through the Earth or a measurement made along an interior path. The value of c is exact in the International System of Units [6].

4 The algebraic bridge

Substituting Eqs. (3) and (4) gives

$$f_{\mathrm{S}}^2 t_R = \frac{1}{4\pi^2}\frac{g_{\mathrm{N}}}{R}\frac{R}{c} = \frac{g_{\mathrm{N}}}{4\pi^2 c}, \tag{5}$$

$$4\pi^2 f_{\mathrm{S}}^2 t_R = \frac{g_{\mathrm{N}}}{c}. \tag{6}$$

The first product has units of inverse time. Multiplication by $4\pi^2$ recovers the acceleration-over-light-speed scale g_{N}/c. In angular-frequency notation the same identity is shorter:

$$\omega_{\mathrm{S}}^2 t_R = \frac{g_{\mathrm{N}}}{c}. \tag{7}$$

Thus g_{N}/c is a derived inverse-time scale; it is not an oscillation frequency of the navigation system.

Multiplying once more by the radius light-time produces a dimensionless result:

$$4\pi^2 f_{\mathrm{S}}^2 t_R^2 = \frac{g_{\mathrm{N}} R}{c^2} = \frac{GM}{Rc^2} \equiv x_R. \tag{8}$$

Here x_R is the compactness of the idealized spherical source. Equations (5)–(8) are exact algebraic identities for the definitions in this Newtonian model. They do not by themselves describe a relativistic correction to the Schuler dynamics.

5 Connection to a first-order static clock rate

For a static clock at radius R outside a spherical source, the exact Schwarzschild lapse is

$$\alpha(R) = \frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{1 - \frac{2GM}{Rc^2}}. \tag{9}$$

The inverse rate and its excess over unity are

$$\delta\tau(R) \equiv \frac{\mathrm{d}t}{\mathrm{d}\tau} = \frac{1}{\alpha(R)}, \qquad \epsilon_{\mathrm{grav}}(R) \equiv \delta\tau(R) - 1. \tag{10}$$

The symbol $\delta\tau$ denotes a dimensionless rate ratio, not an elapsed proper-time interval. For $x_R \ll 1$, the first-order approximation is

$$\epsilon_{\mathrm{grav}}(R) = \frac{1}{\sqrt{1-2x_R}} - 1 \simeq x_R = \frac{g_{\mathrm{N}} R}{c^2} = 4\pi^2 f_{\mathrm{S}}^2 t_R^2. \tag{11}$$

Only the final comparison with ϵ_{grav} is approximate. The preceding relation between the Schuler definitions and x_R is the algebraic identity in Eq. (8). Higher-order clock-rate terms are outside the scope of this first-order treatment.

The shared ratio depends on the specified static reference. The surface clock does not locally register an absolute deficit; ϵ_{grav} compares its rate with t normalized at infinity. The bridge therefore relates a navigation timescale and a chosen clock-rate comparison without turning either into a new physical field.

6 Numerical illustration

For the series' ideal spherical-Earth reference set, take $GM = 3.98600435507 \times 10^{14}\ \mathrm{m^3\,s^{-2}}$, set the spherical model radius $R = 6.371 \times 10^6$ m, and use the exact SI value of c [4, 5, 6]. The model acceleration is derived from GM/R^2, rather than inserted as a separate standard-gravity value. Where a manuscript uses a different Earth radius or gravity convention, that choice should be stated explicitly. The resulting scales, rounded, are shown in Table 1.

Table 1: Schuler, light-time, and first-order clock-rate scales for the chosen ideal Earth.

Quantity	Value
$g_{\mathrm{N}} = GM/R^2$	$9.8203\ \mathrm{m\,s^{-2}}$
Schuler frequency f_{S}	197.6 μHz
Schuler period T_{S}	84.35 min
Radius light-time $t_R = R/c$	21.251 ms
$f_{\mathrm{S}}^2 t_R$	0.8297 nHz
g_{N}/c	32.76 nHz
$x_R = GM/(Rc^2)$	0.6961 ppb

The first-order clock comparison can also be expressed as an accumulated time difference. Over 86 400 s of reference time at infinity, the static surface clock accumulates approximately

$$\Delta t - \Delta\tau \simeq x_R\,(86\,400\ \mathrm{s}) \simeq 60.1\ \mu\mathrm{s}. \tag{12}$$

Over one Schuler period the corresponding first-order difference is about 3.52 μs. These are comparisons of elapsed intervals for the idealized reference and surface clock, not errors in the operation of either clock.

The standard acceleration of gravity, $g_\mathrm{n} = 9.80665\ \mathrm{m\,s^{-2}}$, is a conventional reference value [7]. Substituting it for g_N while keeping the same R gives approximately 0.695 ppb for $g_\mathrm{n}R/c^2$, rather than the 0.696 ppb obtained from $GM/(Rc^2)$ above. The values differ because they represent different choices of input; g_n is not the Newtonian field of this nonrotating spherical mass model. The two should not be mixed within one calculation.

7 Interpretation and limits

The Schuler frequency is a navigation-system error-mode scale. The radius light-time is a way to express a length in time units. Their products yield g_N/c and $g_\mathrm{N}R/c^2$ by algebra. The latter happens to equal the leading weak-field inverse-rate excess for the static surface clock when the same spherical mass model supplies g_N. This correspondence is not evidence that the Schuler oscillation drives, measures, or causes gravitational time dilation.

The numerical example assumes a Newtonian Schuler model and a first-order clock-rate comparison in an ideal, nonrotating spherical exterior. It is not a model of a real terrestrial clock network. Earth's rotation and nonspherical gravity field, altitude, reference-frame conventions, and external sources must be included for practical geodetic or navigation applications. Nor does the frequency-scale product $f_\mathrm{S}^2 t_R$ imply an independently observable oscillation at the nanohertz value.

8 Conclusion

For the ideal spherical Earth, $f_\mathrm{S} = (2\pi)^{-1}\sqrt{g_\mathrm{N}/R}$ and $t_R = R/c$ give the identities $4\pi^2 f_\mathrm{S}^2 t_R = g_\mathrm{N}/c$ and $4\pi^2 f_\mathrm{S}^2 t_R^2 = g_\mathrm{N}R/c^2$. With $g_\mathrm{N} = GM/R^2$, the second quantity is the compactness $GM/(Rc^2)$ and equals the first-order excess of the static inverse clock-rate ratio relative to infinity. Keeping the 2π convention and the approximation order explicit makes the bridge useful without overstating its physical meaning: it is an algebraic connection among established scales, not a new law or a causal mechanism.

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